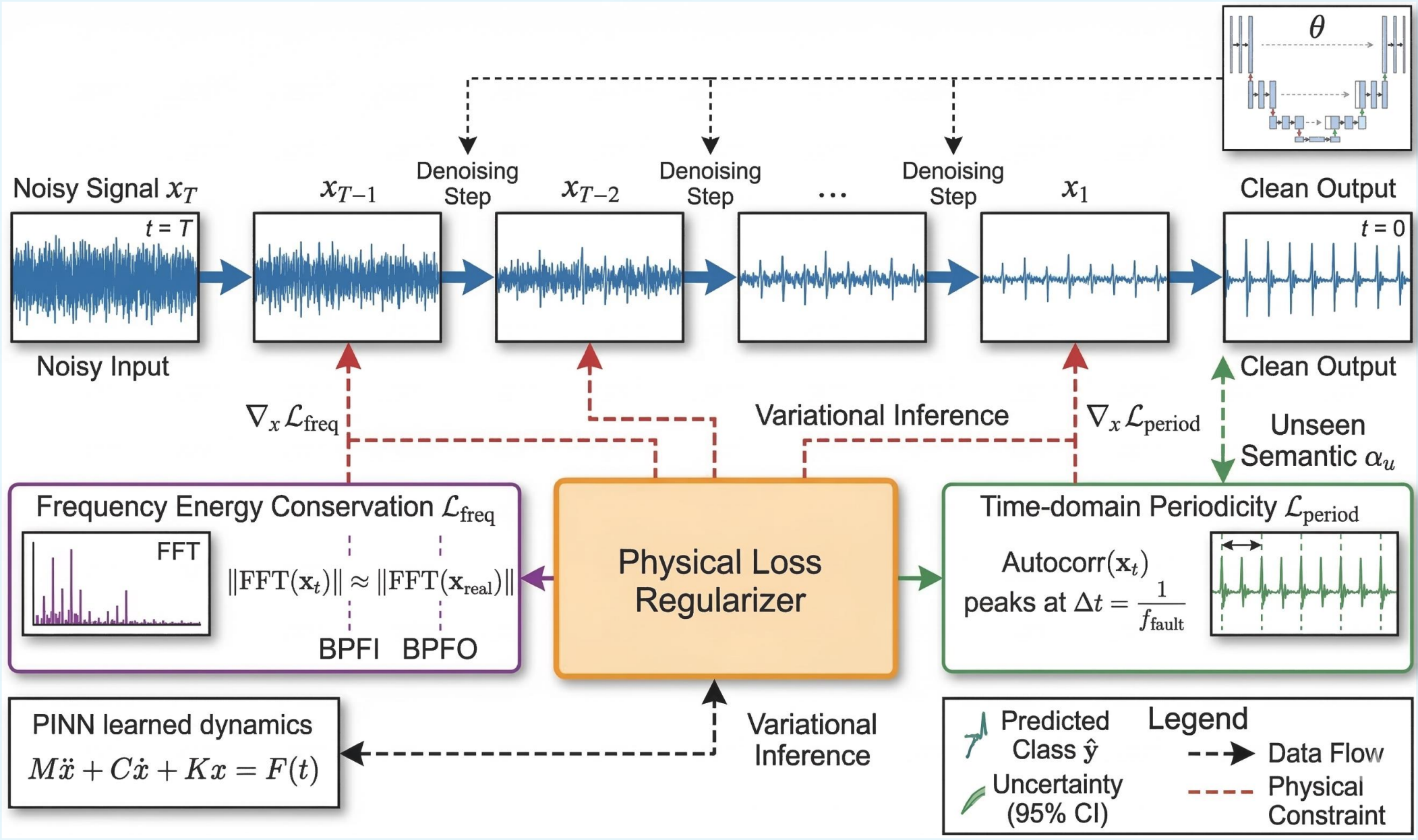
θ
Noisy Signal x_T
x_{T-1}
Denoising Step
x_{T-2}
Denoising Step
...
Denoising Step
x_1
Clean Output
t = T
t = 0
Noisy Input
Clean Output
$\nabla_x \mathcal{L}_{\text{freq}}$
Variational Inference
$\nabla_x \mathcal{L}_{\text{period}}$
Unseen Semantic α_u
Frequency Energy Conservation $\mathcal{L}_{\text{freq}}$
FFT
$\|\text{FFT}(\mathbf{x}_t)\| \approx \|\text{FFT}(\mathbf{x}_{\text{real}})\|$
BPFI
BPFO
Physical Loss Regularizer
Time-domain Periodicity $\mathcal{L}_{\text{period}}$
Autocorr$(\mathbf{x}_t)$
peaks at $\Delta t = \frac{1}{f_{\text{fault}}}$
PINN learned dynamics
$M\ddot{x} + C\dot{x} + Kx = F(t)$
Variational Inference
Legend
Predicted Class $\hat{\mathbf{y}}$
Uncertainty (95% CI)
Data Flow
Physical Constraint